

Introduction to Data Science

Passing the subject:

- hand-in assignments
- final exam in January
 - computer based
 - 20/40 points
 - four additional programming assignments, each 24 points, you can get up to 6 points on the main exam for solving them

Lecture 1

Definition: An experiment is an activity that produces a distinct outcome.

- Ω - sample space that contains all possible outcomes

Ex: Tossing a coin

$$\Omega = H, T$$

Event: a set of outcomes

An event happens after an experiment if the outcome $\omega \in \Omega$; $\omega \in A$

What is probability?

(Idea) (Long term relative frequency) Let's repeat an experiment n -times and count how many times an event A happens

$$N(A, n) = \frac{\text{number of times } A \text{ happens}}{n}$$

- assume the experiment is independent

Example: if $N(A, n) \rightarrow \frac{1}{2}$ as $n \rightarrow \infty$ then we could say "50%" chance of happening probability

Rules

1. $N(\Omega, n) = 1$ - something happens
2. Take two events, A, B . $A \cap B = \emptyset$, then $N(A \cup B, n) = N(A, n) + N(B, n)$

Definition: A function $\mathbb{P}$ that assigns real values to each event is a probability "distribution"/"measure"

if $0 \leq \mathbb{P} \leq 1$

1. $\mathbb{P}(\Omega) = 1$
2. A, B $A \cap B = \emptyset$, $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$

some other notations and cases:

- A and A' (complement) $\mathbb{P}(A) + \mathbb{P}(A') = 1$
- A and B intersect $\mathbb{P}(A \cap B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cup B)$

Bolles inequality (union bound):

$$\mathbb{P}(A \cup B) \leq \mathbb{P}(A) + \mathbb{P}(B)$$

Conditional probability

What is the probability of A , given that I know B happens?

- B becomes the new Ω

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$$

Example: SMS messages

- A - "message is spam"
- B - "free appears in the text"

Law of total probability

Let $A_1, \dots, A_n$, $A_i \cap A_j = \emptyset$

$$\bigcup_{i=1}^n A_i = \Omega$$

then for any $B \in \Omega$

$$\mathbb{P}(B) = \sum_{i=1}^n \mathbb{P}(B|A_i)\mathbb{P}(A_i) = \sum_{i=1}^n \mathbb{P}(B \cap A_i)$$

basically to get probability of B, sum its intersections with A (A makes up the whole omega space)

Lecture 2

Random variables

A random variable (RV) is a mapping

$$X : \Omega \rightarrow \mathbb{R}$$

that assigns a real number $X(\omega)$ to each outcome ω .

$$\mathbb{P}(X \leq 1) := \mathbb{P}(\omega \in \Omega : X(\omega) \leq 1)$$

all outcomes such that $X(\omega) \leq 1$

Example.

$\Omega = \{\text{all possible texts of length 100}\}$

Experiment: Receiving a text of length ≤ 100 ω

$X(u) = 1$ if omega is spam, 0 if not

$$X(u) = \begin{cases} 1 & \text{if } \omega \text{ is spam} \\ 0 & \text{if not} \end{cases}$$

$$P(X = 1) = P(\omega \text{ is spam})$$

Cumulative distribution function CDF

$$F_X : \mathbb{R} \rightarrow [0, 1] \quad F_X(x) = P(X \leq x)$$

Properties of CDF:

- non decreasing $x_1 < x_2$ implies that $F_X(x_1) \leq F_X(x_2)$
- F is normalized $\lim_{x \rightarrow -\infty} F_X(x) = 0$ and $\lim_{x \rightarrow \infty} F_X(x) = 1$
- F is right continuous
- CDF gives the probability that an RV X will take a value less than or equal to a specific value x

If X is discrete, we can define a **probability mass function** (PMF)

$$f_X(x) = P(X = x)$$

Connection to the CDF:

$$F_X(x) = \sum_{t \leq x} f_X(t)$$

For a continuous random variable we distinguish a **probability density function** (PDF) $f_X(x)$. It holds that for PDF:

$$F_X(x) = \int_{-\infty}^x f_X(t)dt \text{ and } f_X(x) = F'_X(x)$$

- PDF describes the relative likelihood for a continuous random variable to take on a specific value
- this is not a probability, it can be greater than 1
- total area under the curve must be 1

Inverse CDF (quantile function)

$$F^{-1}(q) = \inf x : F(x) > q \quad q \in [0, 1]$$

If F is strictly increasing and continuous then $F^{-1}(F(x)) = x$ and $F(F^{-1}(q)) = q$

- Inverse CDF takes probability q and returns the corresponding value of x (an outcome)

We call

- first quartile $F^{-1}(0.25)$
- median/second quartile $F^{-1}(0.5)$
- third quartile $F^{-1}(0.75)$

Transformations of random variables

X is an RV with PDF f_X and CDF F_X . Let $Y = g(X)$ be a transformation of X. How do we compute PDF and CDF of Y?

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \left| \frac{d}{dy} g^{-1}(y) \right|$$

Some important discrete random variables

Point Mass Distribution

$$X \sim \delta_a \text{ if } P(X = a) = 1$$

$$F(X) = \begin{cases} 0 & x < a \\ 1 & x \geq a \end{cases} \quad f(x) = \begin{cases} 1 & \text{if } x = a \\ 0 & \text{else} \end{cases}$$

Discrete Uniform distribution

$$f(x) = \begin{cases} \frac{1}{k} & \text{for } x = 1, \dots, k \\ 0 & \text{otherwise} \end{cases}$$

Bernoulli distribution $P(X=1) = p, P(X=0) = 1-p$

$$X \sim \text{Bernoulli}(p) \quad f(x) = p^x (1-p)^{1-x}$$

Binomial distribution Number of successes in n independent Bernoulli trials with success probability p

$$X \sim \text{Binomial}(n, p) \quad f(x) = \binom{n}{x} p^x (1-p)^{n-x} \text{ for } x = 0, 1, \dots, n$$

x - number of successes

Geometric Distribution

$$X \sim \text{Geom}(p) \quad P(X = k) = p(1-p)^{k-1}, \quad k \geq 1$$

- interpretation: number of trials until first success in independent Bernoulli trials with success probability p

Poisson Distribution

$$X \sim \text{Poisson}(\lambda) \quad f(x) = \frac{\lambda^x e^{-\lambda}}{x!}, \quad x \geq 0$$

Some important continuous random variables

Uniform Distribution

$$X \sim U(a, b) \quad f(x) = \begin{cases} \frac{1}{b-a} & a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

Distribution function is

$$F(x) = \begin{cases} 0 & x < a \\ \frac{x-a}{b-a} & a \leq x \leq b \\ 1 & x > b \end{cases}$$

Normal Gaussian Distribution

$$X \sim N(\mu, \sigma^2) \quad f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp -\frac{1}{2\sigma^2}(x - \mu)^2$$

- standard normal has sigma=1 and mean=0

$$X \sim N(\mu, \sigma^2) \text{ then } Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$$

- sum of gaussian independent RVs has the following distribution

$$\sum X_i \sim N(\sum \mu_i, \sum \sigma_i^2)$$

We can compute any probabilities as long as we compute the CDF $\Phi(z)$ of standard Normal

Exponential distribution

$$X \sim \text{Exp}(\beta) \quad f(x) = \frac{1}{\beta} e^{-x/\beta}, \quad x > 0$$

Lecture 3

Expectation

Toss a dice 100 times

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$$\frac{1 + 3 + 5 + 2 + \dots + 3}{100} = 3.1 \text{ (?) } \$$$

$$\frac{1}{100} * 1 + \frac{2}{100} * 2 + \dots + \frac{6}{100} * 6 = 1 * P(X=1) + 2 * P(X=2) + \dots + 6 * P(X=6)$$

$$E[X] = \sum x \cdot f_X(x) \quad \text{expectation} \quad \text{remainder} \quad f_X(x) - P(X=x) \quad \$$$

What about random vectors? X is R^m and discrete

$$E[X] = \sum x \cdot f_X(x) = \sum_{x_1, x_2, \dots, x_m} x_1 f_X(x_1, x_2, \dots, x_m), \dots, \sum_{x_1, x_2, \dots, x_m} x_m f_X(x_1, x_2, \dots, x_m) \quad \$$$

Linearity of expectation

Say we have X and Y RV's $Z = (X, Y)$

$$E[Z] = E[X + Y] = \sum_{x,y} (x+y) f_{X,Y}(x,y) = \sum_{x,y} x f_{X,Y}(x,y) + \sum_{x,y} y f_{X,Y}(x,y) = E[X] + E[Y] \quad \$$$

Law of the lazy statistician (LOLS)

Let's say we have a random variable X and a function $f: \mathbb{R} \rightarrow \mathbb{R}$

Let $Y = f(X)$

$$E[Y] = \sum_y y f_Y(y) = E[f(X)] = \sum_x f(x) f_X(x)$$

$f_Y(y)$ might be harder to find than $f_X(x)$

Variance

$E[(X - E[X])^2] = E[X^2] - (E[X])^2$ variance is the expectation of the squared deviation from the mean

Mean absolute deviation

$$\mathbb{E}[|X - \mathbb{E}[X]|]$$

If $X_1, X_2, \dots, X_n$ are independent then the variance of the sum is the sum of the variances

Covariance

$$\text{Cov}(X, Y) = \mathbb{E}((X - \mathbb{E}[X])(Y - \mathbb{E}[Y])) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$$

how related are X and Y, how strong the linear relationship is Correlation is normalized covariance $\frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}} \in [-1, 1]$

If X and Y are RVs then

$$V(X + Y) = V(X) + V(Y) + 2\text{Cov}(X, Y)$$

If X and Y are independent then

$$V(X + Y) = V(X) + V(Y)$$

Mean and variance of some important distributions

Distribution	Mean	Variance
Point mass at a	a	0
Bernoulli(p)	p	$p(1 - p)$
Binomial(n, p)	np	$np(1 - p)$
Geometric(p)	$1/p$	$(1 - p)/p^2$
Poisson(λ)	λ	λ
Uniform(a, b)	$(a + b)/2$	$(b - a)^2/12$
Normal(μ, σ^2)	μ	σ^2
Exponential(β)	β	β^2
Gamma(α, β)	$\alpha\beta$	$\alpha\beta^2$
Beta(α, β)	$\alpha/(\alpha + \beta)$	$\alpha\beta/((\alpha + \beta)^2(\alpha + \beta + 1))$
t_ν	0 (if $\nu > 1$)	$\nu/(\nu - 2)$ (if $\nu > 2$)
χ_p^2	p	$2p$
Multinomial(n, p)	np	skibidi
Multivariate Normal(μ, Σ)	μ	Σ

Tower rule

$$\mathbb{E}[\mathbb{E}[X|Y]] = \mathbb{E}[X]$$

Cauchy distribution

$$f_X(x) = \frac{1}{\pi(1 + x^2)}$$

It's mean does not exist because Cauchy has thic tails and hence extreme observations are common

Moments of random variables

The k-th moment of a random variable X is defined as

$$\mu'_k = \mathbb{E}[X^k]$$

The k-th central moment of a random variable X is defined as

$$\mu_k = \mathbb{E}[(X - \mathbb{E}[X])^k]$$

Moment generating function (MGF)

$$M_X(t) = \mathbb{E}[e^{tX}] = \sum_x e^{tx} f_X(x)$$

We obtain the moments of X by differentiating MGF and evaluating at t=0

$$\mu'_k = M_X^{(k)}(0)$$

Notable moments

- 1st moment = expectation
- 2nd central moment = variance
- 3rd central moment = skewness - asymmetry of the distribution
- 4th central moment = kurtosis - "tailedness" of the distribution, measure of the heaviness of the tails, e.g.
 - normal distribution kurtosis = 3
 - Cauchy distribution kurtosis = infinity
 - exponential distribution kurtosis = 9
 - the higher the kurtosis, the heavier the tails -> more prone to outliers and slower convergence to the mean

If kth moment exists and if j < k then the jth moment also exists

If $X_1, \dots, X_n$ are random variables then we define the sample mean and sample variance as:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \quad S_n^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2$$

If $X_1, \dots, X_n$ are iid with mean μ and variance σ^2 then

$$\mathbb{E}[\bar{X}_n] = \mu \quad \mathbb{V}[\bar{X}_n] = \frac{\sigma^2}{n} \quad \mathbb{E}[S_n^2] = \sigma^2$$

Lecture 4

Markov inequality

X - non negative RV,

$$P(X \geq \epsilon) \leq \frac{\mathbb{E}[X]}{\epsilon} \quad \epsilon > 0$$

Chebyshev inequality

$$P(|X - \mathbb{E}[X]| \geq \epsilon) \leq \frac{\mathbb{V}[X]}{\epsilon^2}$$

- requires finite variance
- polynomial decay

Hoeffding inequality

Let $X_1, X_2, \dots, X_n$ be independent RVs such that $a_i \leq X_i \leq b_i$ almost surely. Let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ be the sample mean. Then for any $\epsilon > 0$ it holds that

$$P(|\bar{X}_n - \mathbb{E}[\bar{X}_n]| \geq \epsilon) \leq 2 \exp\left(-\frac{2n\epsilon^2}{\sum_{i=1}^n (b_i - a_i)^2}\right)$$

confidence intervals:

Set the RHS equal to δ and solve for ϵ :

- Hoeffding:

$$2e^{-2n\epsilon^2/(b-a)^2} = \delta \Rightarrow \epsilon = \frac{b-a}{\sqrt{2n}} \sqrt{\ln\left(\frac{2}{\delta}\right)}.$$

OR equivalently

$$\epsilon = \sqrt{\frac{(b-a)^2 \log(2/(1-\alpha))}{2n}}$$

alpha is the confidence level, here 0.95, so inside epsilon we use 1-alpha

- Chebyshev:

$$\frac{\sigma^2}{n\epsilon^2} = \delta \Rightarrow \epsilon = \sqrt{\frac{\sigma^2}{n\delta}}.$$

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Sub-Gaussian and Sub-Exponential random variables

Let X be RV. We call it sub-Gaussian with parameter λ if

$$E[e^{t(X-E[X])}] \leq e^{\frac{\lambda^2 t^2}{2}} \text{ for all } t \in \mathbb{R}$$

We call it sub-Exponential with parameters (ν, b) if

$$E[e^{t(X-E[X])}] \leq e^{\frac{\nu^2 t^2}{2}} \text{ for } |t| < \frac{1}{b}$$

- here we bound the moment generating function
- SG RVs have tails that decay at least as fast as Gaussian tails
- SE tail decays slower than Gaussian but still exponentially
- every SG is SE but not vice versa
- if X is SG then X^2 is SE
- bounded RVs are sub-Gaussian
- sum of SG RVs is SG

Types of convergence of random variables

- convergence in probability

$$X_n \xrightarrow{P} X \text{ if for any } \epsilon > 0 \quad P(|X_n - X| \geq \epsilon) \rightarrow 0 \text{ as } n \rightarrow \infty$$

- convergence almost surely

$$X_n \xrightarrow{a.s.} X \text{ if } P(\lim_{n \rightarrow \infty} X_n = X) = 1$$

- strongest form of convergence
- X_n converges to X for almost every outcome ω in the sample space
- convergence in distribution

$$X_n \xrightarrow{d} X \text{ if } F_{X_n}(x) \rightarrow F_X(x) \text{ for all points } x \text{ where } F_X \text{ is continuous}$$

- sequence of CDFs approaches the limit's CDF (the shape becomes the same)
- convergence in L_p

$$\lim_{n \rightarrow \infty} E[|X_n - X|^p] = 0$$

Hierarchy of implications:

- almost surely $\Rightarrow$ probability $\Rightarrow$ distribution
- $L_p \Rightarrow$ probability

Weak law of large numbers (WLLN)

Let $X_1, X_2, \dots, X_n$ be iid RVs with mean μ and variance $\sigma^2 < \infty$. Let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ be the sample mean. Then for any $\epsilon > 0$ it holds that

$$\mathbb{P}(\bar{X}_n \rightarrow \mu \text{ as } n \rightarrow \infty) = \lim_{n \rightarrow \infty} \mathbb{P}(|\bar{X}_n - \mu| \geq \epsilon) = 0$$

- requires only finite variance

Strong law of large numbers

Let $X_1, X_2, \dots, X_n$ be iid RVs with mean μ . Let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ be the sample mean. Then it holds that $\bar{X}_n \xrightarrow{a.s.} \mu$ as $n \rightarrow \infty$

- requires finite mean (first moment)

Central limit theorem (CLT)

Let $X_1, X_2, \dots, X_n$ be iid RVs with mean μ and variance $\sigma^2 < \infty$. Let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ be the sample mean. Then it holds that $Z_n = \frac{\bar{X}_n - E[\bar{X}_n]}{\sqrt{\text{Var}(\bar{X}_n)}} \xrightarrow{d} N(0, 1)$ as $n \rightarrow \infty$

Lesson 2

Lecture 5 - Estimation

Bias

$$\text{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta$$

MSE

$$\text{MSE} = \mathbb{E}[(\hat{\theta} - \theta)^2] = V[\hat{\theta}] + \text{Bias}(\hat{\theta})^2$$

DKW Inequality

Let $X_1, X_2, \dots, X_n$ be iid RVs with CDF F . Let F_n be the empirical CDF. Then for any $\epsilon > 0$ it holds that

$$\mathbb{P}(\sup_x |F_n(x) - F(x)| > \epsilon) \leq 2e^{-2n\epsilon^2}$$

A non-parametric 1-alpha confidence band for F

$$L(x) = \hat{F}(x) - \epsilon, U(x) = \hat{F}(x) + \epsilon \text{ where } \epsilon = \sqrt{\frac{1}{2n} \ln\left(\frac{2}{\alpha}\right)}$$

Lecture 6 - Random variable generation

Congruential generator

Parameters:

- a - multiplier
- b - increment
- m - modulus

$$D(x) = (ax + b) \mod m$$

Hull-Dobell theorem

Congruential generator (a, b, M) has period M iff:

- $\gcd(b, M) = 1$
- p divides a-1 for every prime p that divides M
- 4 divides a-1 if 4 divides M

Examples of (a,b,M):

- (1103515245,12345,2**32)
- (29,12345,2**32)

Accept-Reject sampler

To sample from a distribution with PDF $f(x)$:

1. Choose a proposal distribution with PDF $g(x)$ such that $f(x) \leq c \cdot g(x)$ for some $c > 0$
2. Repeat:
3. Sample Y from $g(x)$
4. Sample U from $\text{Uniform}(0,1)$
5. If $U \leq f(Y)/(c \cdot g(Y))$, return Y ; else, go back to step 3
6. The accepted samples follow the distribution with PDF $f(x)$

Bax-Muller

To generate standard normal RVs:

1. Generate $U_1, U_2 \sim \text{Uniform}(0,1)$
2. Compute:

$$Z_0 = \sqrt{-2\ln(U_1)} \cos(2\pi U_2) \quad Z_1 = \sqrt{-2\ln(U_1)} \sin(2\pi U_2)$$

3. Z_0 and Z_1 are independent standard normal RVs

Inverse transform sampling

Goal: Generate random samples from a target distribution X using a source of uniform random numbers $U \sim \text{Unif}(0,1)$.

Prerequisites:

1. You must know the Cumulative Distribution Function (CDF), $F(x)$.
2. You must be able to calculate the inverse function, $F^{-1}(u)$.

The Algorithm

1. **Generate** a random number u from the uniform distribution $[0,1]$.
2. **Compute** $x = F^{-1}(u)$.
3. **Result:** The value x is a random draw from the distribution defined by F .

Mathematical Proof

Let $U \sim \text{Unif}(0,1)$. We define $X = F^{-1}(U)$. We verify the CDF of X :

$$P(X \leq x) = P(F^{-1}(U) \leq x) = P(U \leq F(x)) \quad (\text{Applying } F \text{ to both sides}) = F(x) \quad (\text{Property of Uniform distribution})$$

Conclusion: The variable X has the CDF $F(x)$, meaning it follows the desired distribution.

Example: Exponential Distribution

- **PDF:** $f(x) = \lambda e^{-\lambda x}$
- **CDF:** $F(x) = 1 - e^{-\lambda x}$
- **Inverse Step:** Let $u = 1 - e^{-\lambda x}$

$$e^{-\lambda x} = 1 - u$$

$$-\lambda x = \ln(1 - u)$$

$$x = -\frac{1}{\lambda} \ln(1 - u)$$

- **Sampling Formula:** Generate u , then compute $x = -\frac{\ln(1-u)}{\lambda}$.

Lecture 7 - Markov chains

1. Definitions

Irreducible

A Markov chain is **irreducible** if it is possible to reach any state from any other state eventually.

- **Graph view:** The state transition graph is strongly connected.
- **Meaning:** The system implies no "trapping sets" or disjoint parts; everything mixes.

Aperiodic

A Markov chain is **aperiodic** if transitions are not locked into a fixed cyclic rhythm.

- **Definition:** The greatest common divisor (GCD) of the lengths of all paths returning to a state is 1.
- **Rule of thumb:** If a state has a "self-loop" (probability of staying in the same state > 0), the chain is aperiodic.

Reversible

A Markov chain is **reversible** if the flux from $i \rightarrow j$ equals the flux from $j \rightarrow i$ once the system is in equilibrium.

- **Detailed Balance Equation:** $\pi_i P_{ij} = \pi_j P_{ji}$
 - **Intuition:** The process is statistically indistinguishable whether time runs forward or backward.
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2. Relationships

- **Do they exclude each other?** No. A chain can be any combination (e.g., Irreducible and Periodic, or Reversible and Reducible).
 - **Do they imply each other?** Mostly no, they are independent properties.
 - *Note: **Reversibility implies Stationarity.*** If a distribution satisfies detailed balance, it is stationary.
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3. Existence of Stationary Distribution (π)

Condition: A probability distribution π such that $\pi P = \pi$.

Finite State Space

- **General Rule:** At least one stationary distribution always exists.
- **If Irreducible:** The stationary distribution is **unique** and positive.

Infinite State Space

- Existence is not guaranteed.
 - A stationary distribution exists if and only if the chain is **Positive Recurrent** (expected return time to a state is finite).
 - If the chain is *Transient* or *Null Recurrent*, π does not exist (probabilities "leak" out to infinity).
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4. Convergence (The Ergodic Theorem)

For a finite Markov chain to **converge** to its stationary distribution over time (regardless of start state), it must be:

1. **Irreducible**
2. **Aperiodic**

$$\lim_{n \rightarrow \infty} P_{ij}^n = \pi_j$$

Lecture 8 - High dimension

Gaussian Annulus Theorem

In high dimension the Gaussian concentrates in an annulus of radius $\sqrt{d}$.

Johnson - Lindenstrauss random projection theorem:

The Johnson-Lindenstrauss Lemma states that you can project points from a high-dimensional space into a lower-dimensional space while preserving the Euclidean distances between them with high probability.

Here is the breakdown of the implementation steps:

1. The Transformation Matrix (R)

To go from d dimensions to k dimensions, we need a linear transformation. In linear algebra, this is represented by a matrix with d rows (to match the input) and k columns (to produce the output). We fill this matrix with random numbers drawn from a Gaussian (Normal) distribution ($\mu=0, \sigma=1$).

2. The Projection ($X \cdot R$)

We perform matrix multiplication between your input data X (shape n_samples, d) and the random matrix R (shape d, k).

The inner dimensions (d) cancel out.

The resulting shape is (n_samples, k).

3. The Scaling (k1)

This is the most strictly mathematical part. If you just sum up k random Gaussian variables squared, the resulting magnitude grows as k increases. To make sure the distances in the new lower-dimensional space resemble the distances in the original space, we must normalize the result. The scaling factor for Gaussian random projections is k1.

Other notes

Train-Validation-Test splitting

If we want to achieve 60/15/25 split:

1. First get 40% of the data as a 'temp' set and 60% as train set
2. Then split the 'temp' set into validation and test sets according to the formula

$$test_size = \frac{P_{test}}{P_{test} + P_{val}}$$

Bennett's inequality

Let $X_1, X_2, \dots, X_n$ be independent, centered (mean 0) sub-Exponential RVs with parameters (ν_i, b_i) . Let $S_n = \sum X_i$. Then for any $t > 0$ it holds that

$$P(S_n \geq t) \leq \exp\left(-\frac{n\nu_n^2}{b_n^2} h\left(\frac{b_n t}{n\nu_n^2}\right)\right)$$

where

$$\nu_n^2 = \frac{1}{n} \sum_{i=1}^n \nu_i^2, \quad b_n = \max_i b_i, \quad h(u) = (1+u) \ln(1+u) - u$$

Bernstein's inequality is a simpler version of Bennett's inequality

$$P(S_n \geq t) \leq \exp\left(-\frac{t^2/2}{n\nu_n^2 + b_n t/3}\right)$$

Epsilon confidence interval:

$$\varepsilon = \sqrt{\frac{2\sigma^2}{n} \ln\left(\frac{2}{\delta}\right)} + \frac{M}{3n} \ln\left(\frac{2}{\delta}\right)$$

σ^2 - average of variances of X_i M = maximum possible range/bound of X_i δ - confidence level (e.g. 0.01)

Exam 2022 concentration of measure

1. Theoretical Background

Concentration of measure defines how fast a random variable (like an empirical mean) converges to its true expected value as the sample size n grows.

Exponential Concentration (Strong)

The probability of a deviation drops extremely fast (exponentially) as n increases.

$$P(|Z - \mathbb{E}[Z]| \geq \epsilon) \leq C_1 e^{-C_2 n \epsilon^2}$$

This requires the underlying data to have "**light tails**" (extreme values are very rare).

- **Sub-Gaussian:** Tails decay like a Gaussian (e^{-x^2}). Examples: Normal distribution, Bounded variables (Bernoulli, Uniform).
- **Sub-Exponential:** Tails decay like an Exponential ($e^{-|x|}$). Examples: Exponential distribution, Chi-squared.

Polynomial Concentration (Weak)

The probability drops much slower (polynomially), usually driven by **Chebyshev's Inequality**.

$$P(|Z - \mathbb{E}[Z]| \geq \epsilon) \leq \frac{\text{Var}(Z)}{\epsilon^2} \propto \frac{1}{n\epsilon^2}$$

This only requires **Finite Variance**. Extreme values are possible, so we cannot guarantee the tight exponential bound.

Part 1: Exponential Concentration

Correct Answers: [1, 2, 5, 9, 10]

These variables have "light tails" or are bounded, allowing for strong bounds like Hoeffding's or Bernstein's inequalities.

- **1. The empirical mean of i.i.d. sub-Gaussian random variables**
 - **Why:** By definition, sums of independent sub-Gaussian variables are sub-Gaussian. We can use **Hoeffding's Inequality**.
 - **2. The empirical mean of i.i.d. sub-Exponential random variables**
 - **Why:** Sums of sub-Exponentials are sub-Exponential. We can use **Bernstein's Inequality**, which gives an exponential bound (specifically a mix of $e^{-n\epsilon^2}$ for small deviations and $e^{-n\epsilon}$ for large ones).
 - **5. The empirical variance of i.i.d. sub-Gaussian random variables**
 - **Why:** The empirical variance depends on the average of X^2 . If X is sub-Gaussian (tail e^{-x^2}), then $Y = X^2$ is **sub-Exponential** (tail e^{-y}). Since we are taking the mean of sub-Exponential variables (X^2), this falls back to Case 2.
 - **9. The empirical mean of i.i.d. deterministic random variables**
 - **Why:** A deterministic variable is a constant. The variance is 0. It equals its mean with probability 1 (infinite concentration).
 - **10. The empirical tenth moment of i.i.d. Bernoulli random variables**
 - **Why:** A Bernoulli variable B takes values 0, 1. B^{10} also takes values 0, 1. All bounded variables are **sub-Gaussian**, so their mean concentrates exponentially (Hoeffding).
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Part 2: Weak Concentration

Correct Answers: [3, 6, 7, 8]

These variables have tails that are too "heavy" for exponential bounds but still have finite variance, satisfying Chebyshev's inequality.

- **3. The empirical mean of i.i.d. random variables with finite variance**
 - **Why:** We only know the variance is finite. We don't know if higher moments exist or if tails decay exponentially. We can strictly only apply **Chebyshev's Inequality**.
- **6. The empirical variance of i.i.d. sub-Exponential random variables**
 - **Why:** Empirical variance relies on X^2 . If X is sub-Exponential ($e^{-|x|}$), then $Y = X^2$ has a tail that behaves like $e^{-\sqrt{y}}$. This is heavier than a standard exponential tail (e^{-y}). It does *not* satisfy the sub-Exponential definition for Bernstein's bound, but since all moments exist, the variance is finite.
- **7. The empirical third moment of i.i.d. sub-Gaussian random variables**
 - **Why:** We look at X^3 . If X is sub-Gaussian (e^{-x^2}), then $Z = X^3$ has tails like $e^{-z^{2/3}}$. This decay is slower than exponential (e^{-z}), so no Hoeffding/Bernstein. However, finite moments exist.
- **8. The empirical fourth moment of i.i.d. sub-Gaussian random variables**

- **Why:** We look at X^4 . If X is sub-Gaussian (e^{-x^2}), then $W = X^4$ has tails like $e^{-\sqrt{w}}$. This is significantly slower than exponential.

Excluded Case

- **4. The empirical variance of i.i.d. random variables with finite variance**
 - **Why:** To compute the *concentration* of the variance, we need the **variance of the sample variance**, which depends on the **fourth moment** ($E[X^4]$). The problem only guarantees finite variance ($E[X^2] < \infty$), not finite fourth moment.

Steps for Model Calibration

Calibration is the process of adjusting the output probabilities of a classification model so that they reflect the true likelihood of the event occurring. For example, if a model predicts a 70% chance of rain for 10 days, it should actually rain on approximately 7 of those days.

Here are the standard steps to calibrate a model, specifically using the approach required in Problem 2 (isotonic regression or decision trees on a calibration set).

1. Split Your Data

You cannot train and calibrate on the same data because the model is already "biased" toward fitting the training labels. You need three distinct sets:

- **Training Set:** Used to train the base classifier (e.g., your Logistic Regression or **ProportionalSpam** model).
- **Calibration Set:** A held-out subset used *only* to learn the mapping from "raw scores" to "true probabilities."
- **Test Set:** Used to evaluate the final performance.

2. Train the Base Model

Train your classifier (e.g., **problem2_ps**) on the **Training Set**.

- The model learns to distinguish classes (0 vs 1).
- It outputs a score or a probability (e.g., 0.8), but this probability might be uncalibrated (e.g., too confident or not confident enough).

3. Generate Raw Predictions

Use the trained base model to predict scores for the **Calibration Set**.

- Do *not* predict binary labels (0 or 1).
- Predict the continuous scores or probabilities.
- Let's call these predictions P_{calib} .

4. Train the Calibrator

Train a second regression model (the calibrator) to map P_{calib} to the true labels of the calibration set (Y_{calib}).

- **Input (X):** The raw predictions from the base model (P_{calib}). Note: You usually need to reshape this to **(n_samples, 1)**.
- **Target (Y):** The actual ground truth labels (Y_{calib} , which are 0s and 1s).
- **Model:** Common choices are Isotonic Regression (non-parametric, strictly increasing) or a Decision Tree Regressor (as used in this exam).

Conceptually, the calibrator learns: "When the base model says 0.7, the actual empirical probability is 0.6."

5. Calibrate Test Predictions

To make final predictions on new data (the **Test Set**):

1. **Step A:** Use the **Base Model** to predict raw scores on X_{test} .
2. **Step B:** Feed those raw scores into the **Calibrator**.
3. **Step C:** The output of the calibrator is your final, calibrated probability.

6. Evaluate

Check the reliability of your new probabilities using metrics like:

- **Brier Score:** Mean squared difference between predicted probability and actual outcome.
 - **Calibration Plot (Reliability Diagram):** Plots predicted probability vs. observed frequency. Ideally, this should be a diagonal line $y = x$.
 - **0-1 Loss:** (If you need a hard classification) Apply a threshold (usually 0.5) to the calibrated probabilities and calculate error rate.
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Additional notes at: <https://bluej1.github.io/IntroDSExamMaterial/>

solutions to exercises: https://github.com/sajad13901/Statistics_Wasserman/tree/main

<https://michalkamin.github.io/intro-to-ds-study-helper/>